REPORT

BFS (Breadth-First Search)

Aim

To implement Breadth-First Search (BFS) to traverse a graph by visiting vertices level by level using a queue.

Objectives

- To understand the concept of graph traversal.
- To implement BFS using a queue.
- To visit all reachable vertices systematically.
- To understand the FIFO principle of a queue.

Requirements

- Computer/Laptop
- Programming language such as C, C++, Java, or Python
- Code editor/IDE
- Graph with vertices and edges

Theory

Breadth-First Search (BFS) is a graph traversal technique that visits vertices level by level. It starts from a selected vertex and first visits all its adjacent vertices before moving to the next level. BFS uses a **Queue**, which follows the **FIFO (First In, First Out)** principle. A visited list is maintained to avoid visiting the same vertex repeatedly.

Algorithm

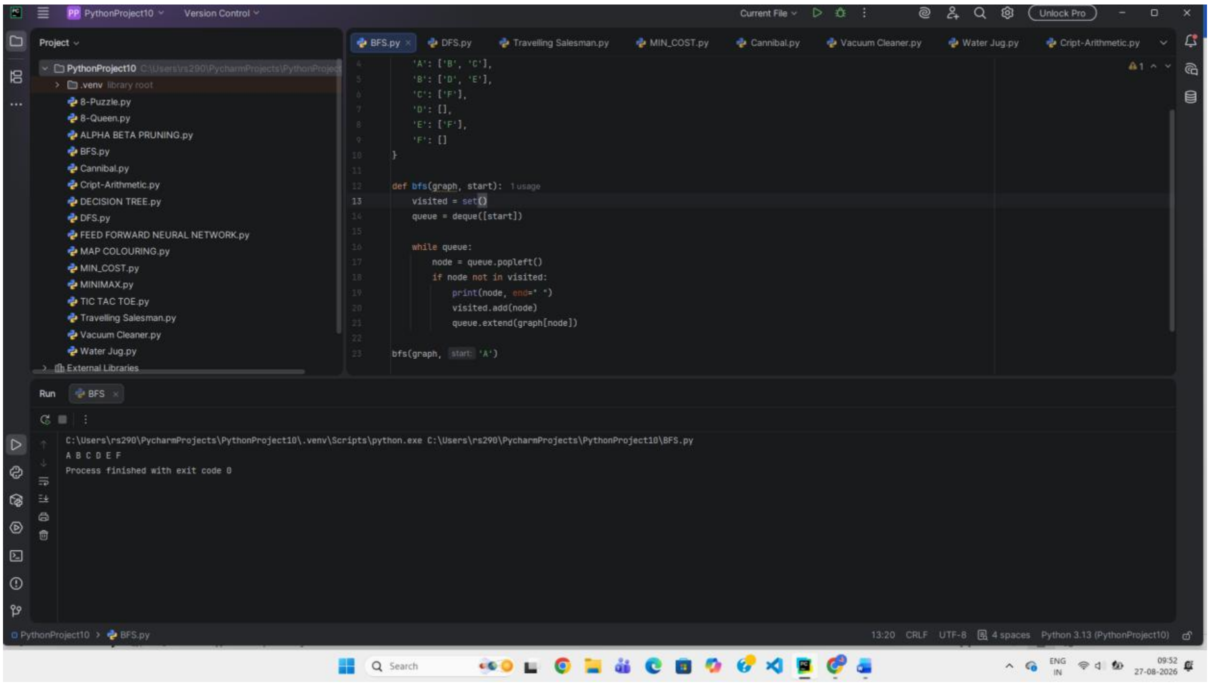
1. Start with the given starting vertex.
2. Mark the starting vertex as visited.
3. Insert the starting vertex into the queue.
4. Remove the front vertex from the queue.
5. Check all adjacent vertices of the removed vertex.
6. Mark each unvisited adjacent vertex as visited.

7. Insert the newly visited vertices into the queue.
8. Repeat steps 4–7 until the queue becomes empty.
9. Display the vertices in the order they were visited.

PROGRAM:

```
from collections import deque

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

def bfs(graph, start):
    visited = set()
    queue = deque([start])

    while queue:
        node = queue.popleft()
        if node not in visited:
            print(node, end=" ")
            visited.add(node)
            queue.extend(graph[node])

bfs(graph, 'A')
```

OUTPUT:



Result

Thus, the **Breadth-First Search (BFS)** traversal of the given graph is successfully performed using a queue, visiting the vertices level by level.